LIBBY — RECOMMENDATIONS

Recommendations — gist-sdh-multifocal-resected-m1-k4n8

Ranked therapeutic options with likelihood, toxicity, mechanism risk, and key references

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Libby is an experimental decision-support tool. The recommendations in this report have not been reviewed by a clinician treating this patient. Do not act on this report without consulting a qualified oncologist.

Recommendations — gist-sdh-multifocal-resected-m1-k4n8

Profile snapshot

- **Primary site:** stomach
- **Histology:** gastrointestinal stromal tumor (GIST), SDH-deficient subtype (KIT/PDGFRA wild-type)
- **Stage:** ypT3(m)N0M1 — multifocal gastric GIST with a single M1 deposit in the ligament of Treitz; R0 distal gastrectomy + Roux-en-Y; 0/7 nodes; largest mass 7.5 cm; M1 deposit completely resected
- **ECOG:** 1
- **Age band:** 40-49
- **Sex:** unknown
- **Biomarkers:** SDHA L306P mutated (VAF 44%; germline-vs-somatic not yet established) (ngs_pending); SDHA Y408N mutated (VAF 41%, somatic); SDHA E350fs mutated (ctDNA) (ngs_pending); PIK3CA R93W mutated (ctDNA) (ngs_pending); MAP2K1 P124S mutated (ctDNA) (ngs_pending); POLE R1679C mutated (germline); CD117 (KIT) protein positive; DOG1 protein positive; Desmin negative; SDHB protein intact; KIT wild-type; PDGFRA wild-type; TMB <1 mut/Mb; MSI status MSS; NTRK no fusion detected; RET no actionable alteration; BRAF wild-type
- **Efficacy/toxicity weight:** 0.65

7 ranked therapeutic options. Biomarker workup steps live in the standalone Target validation paths report.

Dsdh Ihc interventions

1. Sunitinib 50 mg PO 4-on/2-off (or 37.5 mg continuous) at first measurable recurrence — NOT adjuvant *(conditional on dsdh_ihc positive)*

Likelihood of effect: Moderate-to-low in dSDH-GIST: 7/38 (18%) objective response, Boikos 2016 — the best-replicated systemic signal, but well below KIT-mutant activity.

Toxicity burden: Moderate (fatigue, hand-foot skin reaction, hypertension, diarrhea)

Counter-productive MoA:Low (Anti-angiogenic VEGFR blockade can impair wound healing — relevant if a future re-resection is on the table.)

Overall:The one FDA-approved, guideline-preferred contingency with dSDH-specific data (18% ORR); modest but predictable where the newer agents are not.

Key references: PMID 27011036 · PMID 17046465

2. Temozolomide (MGMT-methylation-enriched), trial route preferred (NCT05661643 / NCT03556384) at first measurable recurrence *(conditional on dsdh_ihc positive)*

Likelihood of effect: Moderate only if MGMT-methylated (~24% of dSDH-GIST): 2/5 PR, Yebra 2022; low or absent in unmethylated tumors. Gate on testing.

Toxicity burden: Moderate (thrombocytopenia, neutropenia, fatigue, nausea)

Counter-productive MoA:Low (Alkylator exposure without MGMT methylation incurs cytopenias for little antitumor effect; theoretical hypermutation/resistance risk over long use.)

Overall:Highest reported ORR (40%, n=5) but only in the ~24% MGMT-methylated subset; gate on methylation testing before exposing anyone to alkylator cytopenias.

Key references: PMID 34426440 · PMID 36198483 · NCT05661643

3. Olverembatinib (HQP1351, multikinase inhibitor), trial route only (phase 1b signal; phase-3 NCT06640361) at first measurable recurrence *(conditional on dsdh_ihc positive)*

Likelihood of effect: Moderate: 6/26 (23.1%) PR, mPFS 25.7 mo (Qiu 2025) in the exact subtype, but a single phase-1b cohort, unreplicated.

Toxicity burden: Low (anemia, transaminitis, hyperuricemia, leukocytosis — all G3+ rates < 20%)

Counter-productive MoA:Moderate (Non-canonical MoA in a KIT-WT tumor; efficacy borrows from VEGFR/FGFR/HIF-CD36 axes, not the labeled KIT/ABL target.)

Overall:Cleanest safety sheet and replicated dSDH activity (23.1% ORR, 25.7-mo PFS), but phase 1b, China-sponsored, and its phase-3 entry needs SDHB-IHC loss this patient may not show.

Key references: PMID 41184234 · NCT06640361

4. Rogaratinib (pan-FGFR1-4 inhibitor), trial route only (NCI cohort closed to accrual; FGFR-pathway evidence anchor) at first measurable recurrence *(conditional on dsdh_ihc positive)*

Likelihood of effect: Highest reported: 10/24 (41.7%) PR, mPFS 31 mo (Merriam 2026), but single-arm n=24 and unreplicated; DCR may overstate via indolence.

Toxicity burden: Unresolved (per-grade AE table not retrievable; class hyperphosphatemia, fatigue, diarrhea reported without grades)

Counter-productive MoA:Moderate (Single-arm DCR may reflect dSDH indolence rather than drug effect; the FGFR-pathway dependence is unreplicated.)

Overall:Best ORR yet in dSDH-GIST (41.7%), but single-arm n=24, AE table unretrievable, and the trial is closed to accrual — access, not efficacy, is the wall.

Key references: PMID 42191879 · PMID 31666694 · NCT04595747

5. Belzutifan (HIF-2alpha inhibitor) 120 mg PO daily, off-label or LITESPARK-015 wt-GIST cohort (NCT04924075) at first measurable recurrence *(conditional on dsdh_ihc positive)*

Likelihood of effect: Uncertain in GIST — no readout; cross-indication PPGL cohort ORR 26% (LITESPARK-015) in sibling SDH-loss biology is the only efficacy anchor.

Toxicity burden: Moderate (anemia, fatigue, hypoxia)

Counter-productive MoA:Low (Direct driver-axis blockade, but a mechanism-matched agent (guadecitabine) failed in this subtype — target engagement may not translate.)

Overall:Most direct shot at the HIF-2alpha driver and accessible off-label, but no GIST readout and a mechanism-matched precedent (guadecitabine) already failed.

Key references: NCT04924075 · PMID 35324464 · PMID 15652751

6. Pemigatinib (FGFR1-3 inhibitor), PEMIGIST trial route (NCT07434843, recruiting, DFCI) at first measurable recurrence *(conditional on dsdh_ihc positive)*

Likelihood of effect: Unknown — no clinical readout in dSDH-GIST; the rationale rests on the rogaratinib FGFR signal and a shared PDX response (pmid:31666694).

Toxicity burden: Moderate (hyperphosphatemia, stomatitis, serous retinal detachment, nail/skin changes)

Counter-productive MoA:Low (FGFR-pathway bet with no clinical readout; shares rogaratinib's unreplicated-pathway uncertainty.)

Overall:The open FGFR door — recruiting in-subtype — but riding entirely on the rogaratinib signal and shared preclinical models, with no clinical readout of its own.

Key references: NCT07434843 · PMID 31666694 · PMID 42191879

7. Regorafenib 160 mg PO 3-weeks-on/1-off, guideline-sequenced 3L TKI after sunitinib at recurrence *(conditional on dsdh_ihc positive)*

Likelihood of effect: Low/modest — no dSDH-specific data; GRID HR 0.27 is KIT-mutant, dSDH activity extrapolated (von Mehren 2025).

Toxicity burden: Moderate (hand-foot skin reaction, hypertension, diarrhea)

Counter-productive MoA:Low (Anti-angiogenic multikinase activity; dSDH benefit is extrapolated from a KIT-mutant pivotal, not demonstrated.)

Overall:Guideline-sequenced 3L TKI; the registration evidence is KIT-mutant and the dSDH benefit is extrapolated, not demonstrated.

Key references: PMID 23177515 · PMID 40045030

Evidence in detail

Sunitinib 50 mg PO 4-on/2-off (or 37.5 mg continuous) at first measurable recurrence — NOT adjuvant

Boikos SA et al. *JAMA Oncology* 2016

- **Disease context:KIT/PDGFRA wild-type GIST (NIH Pediatric & Wild-Type GIST Clinic cohort; majority SDH-deficient)** (2L+) — 95 patients with adequate tumor specimens from the NIH wt-GIST clinic; median age 23; 70% female; molecular subtypes: SDH-mutant (n=63), SDH-epimutant (n=21), SDH-competent (n=11). The cohort skews young and pediatric-onset, with gastric primaries dominating the SDH-deficient strata.
- **n:** 95
- **Toxicities:** Toxicity not the focus of this cohort report; sunitinib and imatinib safety as expected for label dosing.
- **Efficacy:ORR** 2%; **ORR** 18%

Demetri GD et al. *Lancet* 2006

- **Disease context:advanced/metastatic GIST after imatinib failure or intolerance** (2L+) — 312 patients (207 sunitinib, 105 placebo) with imatinib-resistant or -intolerant GIST. No SDH-status stratification; the cohort was overwhelmingly KIT-mutant by historical molecular epidemiology, with dSDH-GIST under-represented.
- **n:** 312
- **Toxicities:fatigue** (grade any); **diarrhea** (grade any); **skin discoloration** (grade any); **nausea** (grade any)
- **Efficacy:TTP** 27.3 weeks; **TTP** 6.4 weeks; **HR_PFS** 0.33

Temozolomide (MGMT-methylation-enriched), trial route preferred (NCT05661643 / NCT03556384) at first measurable recurrence

Yebra M et al. *Clinical Cancer Research* 2022

- **Disease context:advanced SDH-mutant GIST (SDHA/SDHB/SDHC/SDHD mutation, biallelic in tumor)** (2L+) — 5 SDH-mutant GIST patients (1 SDHA, 2 SDHB, 1 SDHC, 1 SDHD); median age 29.7; 2F/3M; median 1 prior TKI line (range 0-2). Sicklick / UCSD case series paired with patient-derived xenograft modeling.
- **n:** 5
- **Toxicities:** Paper does not publish a per-AE safety table for the clinical cohort; the translational focus is on the PDX-driven mechanism rationale. Standard TMZ low-dose continuous schedule toxicity profile (cytopenias, fatigue, nausea) presumed.
- **Efficacy:ORR** 40%; **DCR** 100%; **OS** 1.9 years

Giger OT et al. *Journal of Clinical Pathology* 2022

- **Disease context:SDH-deficient and KIT/PDGFRA-WT GIST** (any) — 65 GIST tumors profiled: 47 wt-GIST (33 SDH-deficient), 21 tyrosine-kinase-mutant. Single-institution Cambridge cohort.
- **n:** 65
- **Toxicities:** —
- **Efficacy:biomarker** 24%

Olverembatinib (HQP1351, multikinase inhibitor), trial route only (phase 1b signal; phase-3 NCT06640361) at first measurable recurrence

Qiu HB et al. *Signal Transduction and Targeted Therapy* 2025

- **Disease context:advanced SDH-deficient GIST after prior TKI therapy** (2L+) — 26 SDH-deficient GIST patients within a 66-patient phase 1b that also enrolled other solid tumors; median age 30 (range 13-56), 26.9% male, heavily pretreated with 65.4% having had at least two prior TKIs and 42.3% three.
- **n:** 26

- **Toxicities:treatment-related AE grade >=3 (composite)** (grade >=3): 13.6% (n=66); **leukocyte count increased** (grade any): 48.5% (n=66); **anemia** (grade any): 47.0% (n=66); **anemia** (grade >=3): 6.1% (n=66); **aspartate aminotransferase increased** (grade any): 47.0% (n=66); **aspartate aminotransferase increased** (grade >=3): 3.0% (n=66); **hyperuricemia** (grade any): 47.0% (n=66); **pyrexia** (grade any): 42.4% (n=66); **treatment discontinuation due to AE** (grade any): 0% (n=66)
- **Efficacy:ORR** 23.1% (95% CI 9-43.7); **DCR** 84.6% (95% CI 65.1-95.6); **PFS** 25.7 months; **PFS_rate** 61.5%; **OS_rate** 100%

Rogaratinib (pan-FGFR1-4 inhibitor), trial route only (NCI cohort closed to accrual; FGFR-pathway evidence anchor) at first measurable recurrence

Merriam P et al. *Nature Medicine* 2026

- **Disease context:advanced SDH-deficient GIST (enrolled regardless of FGFR alteration status)** (2L+) — 24 SDH-deficient GIST patients enrolled regardless of FGFR alteration status, on the rationale of FGFR/FGF upregulation in the subtype.
- **n:** 24
- **Toxicities:hyperphosphatemia** (grade any): (n=24); **fatigue** (grade any): (n=24); **diarrhea** (grade any): (n=24)
- **Efficacy:ORR** 41.7%; **DCR** 91.7%; **PFS** 31.0 months; **PFS_rate** 77.4% (95% CI 61.7-97.1)

Pemigatinib (FGFR1-3 inhibitor), PEMIGIST trial route (NCT07434843, recruiting, DFCI) at first measurable recurrence

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- **Disease context:advanced SDH-deficient GIST (enrolled regardless of FGFR alteration status)** (2L+) — 24 SDH-deficient GIST patients enrolled regardless of FGFR alteration status, on the rationale of FGFR/FGF upregulation in the subtype.
- **n:** 24
- **Toxicities:hyperphosphatemia** (grade any): (n=24); **fatigue** (grade any): (n=24); **diarrhea** (grade any): (n=24)
- **Efficacy:ORR** 41.7%; **DCR** 91.7%; **PFS** 31.0 months; **PFS_rate** 77.4% (95% CI 61.7-97.1)

Regorafenib 160 mg PO 3-weeks-on/1-off, guideline-sequenced 3L TKI after sunitinib at recurrence

Demetri GD et al. *Lancet* 2013

- **Disease context:advanced GIST after imatinib + sunitinib failure** (3L+) — 199 patients (133 regorafenib, 66 placebo) with progression after both imatinib and sunitinib. No prospective SDH-status stratification; KIT-mutant disease dominated the population.
- **n:** 199
- **Toxicities:hypertension** (grade >=3): 23% (n=132); **hand-foot skin reaction** (grade >=3): 20% (n=132); **diarrhea** (grade >=3): 5% (n=132)
- **Efficacy:PFS** 4.8 months; **PFS** 0.9 months; **HR_PFS** 0.27 (95% CI 0.19-0.39)